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WO-300-06 — Replace spool SP03 of 4"-P150-CWS-3003

Area	A300 Exchangers and drum	**Task**	SR Spool replacement
Priority	3 (reliability)	**Issued by**	Facility owner (demonstration)
Primary asset	E-102 — E-102 shell-and-tube exchanger 36" ID × 192" — E 134'-6" N -25'-0" EL 106'-6"		
Equipment	E-102	**Lines**	4"-P150-CWS-3003
Isometrics	ISO-CWS-3003 (owner's drawings, attached)		
Permits	General work permit, Hot work permit		

Reason

Internal corrosion on the cooling water supply branch.

Scope of work

1. Isolate per the isolation list; drain, vent and purge 4"-P150-CWS-3003 between the isolations; confirm gas-free for hot work.
2. Remove spool 4"-P150-CWS-3003-SP03 (5 items, 39'-7-5/16" of pipe) — cut at the field welds and break the flanged joints at its ends as listed.
3. Fabricate the replacement spool to ISO-4"-P150-CWS-3003 with the same items (parts list), welding to WS-01; 4 shop welds.
4. Install the spool; make 2 field welds; new gaskets and studs at flanged ends, tightened per SW-WO-5.
5. NDE per the NDE table; hydrostatic test at 210 psi for 30 minutes with the relief valves blinded (SW-TEST-1).
6. Remove blinds, restore the line-up, hand back.

Affected items: 4"-P150-CWS-3003/p8, 4"-P150-CWS-3003/el9, 4"-P150-CWS-3003/p10, 4"-P150-CWS-3003/el11, 4"-P150-CWS-3003/p12

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-1, SW-NDE-2, SW-NDE-4, SW-TEST-1, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-4, SW-WO-5, SW-WO-7, SW-WPS-1, SW-WPS-2, SW-WPS-3

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-CWS-3003	4"-P150-CWS-3003	4"-P150-CWS-3003	1	agrees with the model

ISO-CWS-3003 (PDF): ../isometrics-imported/ISO-CWS-3003.pdf — the sheets that hold the work follow the ITP.

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
E-102	equipment	E-102 shell-and-tube exchanger 36" ID × 192"		E 134'-6" N -25'-0" EL 106'-6"
4"-P150-CWS-3003/p8	item	PIPE 4" P150 BEVELLED ENDS	4"-P150-CWS-3003	E 136'-6" N 0.000" EL 125'-10-3/16"
4"-P150-CWS-3003/el9	item	ELBOW 90° R = 5/3 OD 4" P150	4"-P150-CWS-3003	E 136'-6" N -3-5/16" EL 126'-2-11/16"
4"-P150-CWS-3003/p10	item	PIPE 4" P150 BEVELLED ENDS	4"-P150-CWS-3003	E 136'-6" N -12'-6" EL 126'-6"
4"-P150-CWS-3003/el11	item	ELBOW 90° R = 5/3 OD 4" P150	4"-P150-CWS-3003	E 136'-6" N -24'-8-11/16" EL 126'-2-11/16"
4"-P150-CWS-3003/p12	item	PIPE 4" P150 BEVELLED ENDS	4"-P150-CWS-3003	E 136'-6" N -25'-0" EL 118'-2-3/16"
4"-P150-CWS-3003/A3	item	VALVE, GATE, WEDGE, OS&Y 4" P150	4"-P150-CWS-3003	E 136'-6" N 0.000" EL 124'-7-7/8"
4"-P150-CWS-3003/A6	item	FLANGE WELD-NECK 4" P150 RAISED FACE	4"-P150-CWS-3003	E 136'-6" N 0.000" EL 125'-6-5/16"
4"-P150-CWS-3003/B2	item	VALVE, GATE, WEDGE, OS&Y 4" P150	4"-P150-CWS-3003	E 136'-6" N -25'-0" EL 109'-3-13/16"
4"-P150-CWS-3003/B5	item	FLANGE WELD-NECK 4" P150 RAISED FACE	4"-P150-CWS-3003	E 136'-6" N -25'-0" EL 110'-2-5/16"
4"-P150-CWS-3003/W2	weld	Butt weld 4" P150 (field)	4"-P150-CWS-3003	E 136'-6" N 0.000" EL 125'-9-1/16"
4"-P150-CWS-3003/W3	weld	Butt weld 4" P150 (shop)	4"-P150-CWS-3003	E 136'-6" N 0.000" EL 125'-11-5/16"
4"-P150-CWS-3003/W4	weld	Butt weld 4" P150 (shop)	4"-P150-CWS-3003	E 136'-6" N -6-11/16" EL 126'-6"
4"-P150-CWS-3003/W5	weld	Butt weld 4" P150 (shop)	4"-P150-CWS-3003	E 136'-6" N -24'-5-5/16" EL 126'-6"
4"-P150-CWS-3003/W6	weld	Butt weld 4" P150 (shop)	4"-P150-CWS-3003	E 136'-6" N -25'-0" EL 125'-11-5/16"
4"-P150-CWS-3003/W7	weld	Butt weld 4" P150 (field)	4"-P150-CWS-3003	E 136'-6" N -25'-0" EL 110'-5"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	valve	4"-P150-CWS-3003/A3	4"-P150-CWS-3003	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P150 (4"-P150-CWS-3003/A3)	E 136'-6" N 0.000" EL 124'-7-7/8"		

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
2	blind	4"-P150-CWS-3003/A6	4"-P150-CWS-3003	Fit a line blind between 4"-P150-CWS-3003/A3 and FLANGE WELD-NECK 4" P150 RAISED FACE (4"-P150-CWS-3003/A6), work side	E 136'-6" N 0.000" EL 125'-3-5/8"		
3	valve	4"-P150-CWS-3003/B2	4"-P150-CWS-3003	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P150 (4"-P150-CWS-3003/B2)	E 136'-6" N -25'-0" EL 109'-3-13/16"		
4	blind	4"-P150-CWS-3003/B5	4"-P150-CWS-3003	Fit a line blind between 4"-P150-CWS-3003/B2 and FLANGE WELD-NECK 4" P150 RAISED FACE (4"-P150-CWS-3003/B5), work side	E 136'-6" N -25'-0" EL 109'-11-9/16"		

Parts

Description	Qty	Replaces
PIPE 4" P150 BEVELLED ENDS	3	
ELBOW 90° R = 5/3 OD 4" P150	2	

New welds (SW-WO-4)

Weld	Replaces	On drawing	Line	Size	Wall	Shop/field	Spec	NDE	Location	Welder	Date
WO-300-06-W01	4"-P150-CWS-3003/W2	ISO-CWS-3003 sh 1 W2	4"-P150-CWS-3003	4" P150	0.1875 in	field	WS-01	VT + RT	E 136'-6" N 0.000" EL 125'-9-1/16"		
WO-300-06-W02	4"-P150-CWS-3003/W3	ISO-CWS-3003 sh 1 W3	4"-P150-CWS-3003	4" P150	0.1875 in	shop	WS-01	VT	E 136'-6" N 0.000" EL 125'-11-5/16"		
WO-300-06-W03	4"-P150-CWS-3003/W4	ISO-CWS-3003 sh 1 W4	4"-P150-CWS-3003	4" P150	0.1875 in	shop	WS-01	VT	E 136'-6" N -6-11/16" EL 126'-6"		
WO-300-06-W04	4"-P150-CWS-3003/W5	ISO-CWS-3003 sh 1 W5	4"-P150-CWS-3003	4" P150	0.1875 in	shop	WS-01	VT	E 136'-6" N -24'-5-5/16" EL 126'-6"		
WO-300-06-W05	4"-P150-CWS-3003/W6	ISO-CWS-3003 sh 1 W6	4"-P150-CWS-3003	4" P150	0.1875 in	shop	WS-01	VT	E 136'-6" N -25'-0" EL 125'-11-5/16"		
WO-300-06-W06	4"-P150-CWS-3003/W7	ISO-CWS-3003 sh 1 W7	4"-P150-CWS-3003	4" P150	0.1875 in	field	WS-01	VT	E 136'-6" N -25'-0" EL 110'-5"		

Welding specification WS-01 — Butt welds, pipe grade, wall to 1/2 in

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Scope	Pipe, fittings and weld-neck flanges of Grade 35 pipe / Grade 36 forging, wall up to 1/2 in
Joint	Single-V, 60° included, root opening 1/8 in, root face 1/16 in (SW-WD-4)
Process	Root: gas-shielded tungsten arc. Fill and cap: shielded metal arc.
Filler	Root SW-F1 wire; fill and cap SW-F2 covered electrode (matching strength, low hydrogen)
Preheat	≥ 30 °F above dew point and ≥ 60 °F, 3 in each side
Interpass	≤ 450 °F
Post-weld heat treatment	Not specified (owner to state)
Rules	SW-WPS-1, SW-WPS-2, SW-WPS-3, SW-WD-4

NDE

Method	Extent	Welds	Basis
VT	100 % of new welds, before and after welding	WO-300-06-W01, WO-300-06-W02, WO-300-06-W03, WO-300-06-W04, WO-300-06-W05, WO-300-06-W06	SW-NDE-1
RT (UT where RT is impractical)	10 % of new welds, at least one per welder (class W150); welds shown are preliminary, final choice by the owner's inspector	WO-300-06-W01	SW-NDE-2

Test — Hydrostatic test

Description	Test boundary between the isolations; relief valves blinded or removed
Pressure	210 psi
Hold	30 minutes, no leak, no pressure drop
Basis	SW-TEST-1

Inspection and test plan (SW-ITP-1..3)

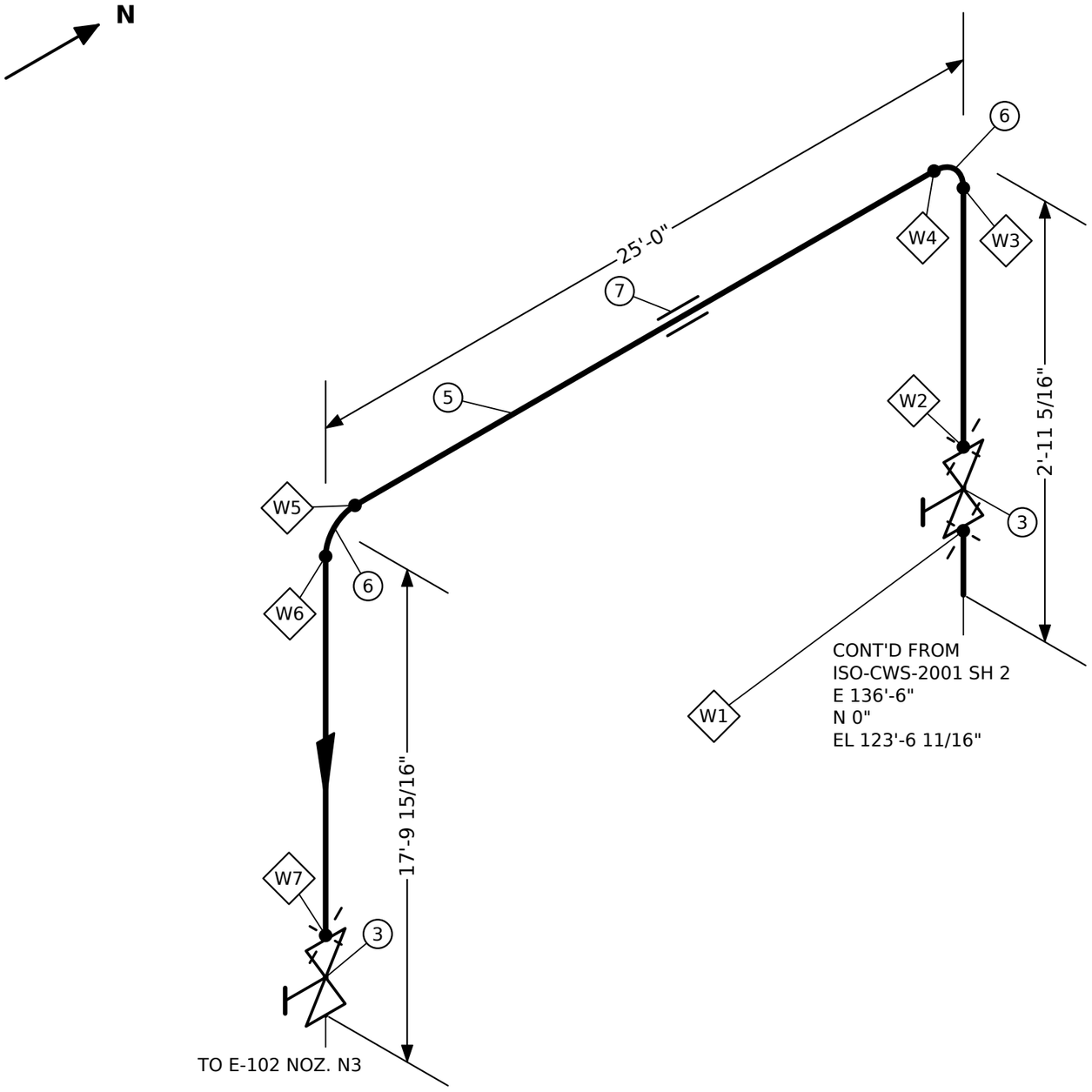
H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Line drained, vented and purged; gas-free for hot work	Owner hot-work procedure	Gas test within owner limits	Gas test record, hot work permit	H	H	—				
3	Replacement materials verified against the parts list (size, class, grade, heat marking)	Parts list, SW-LC-3	All items match the line class	Material receiving record	R	R	R				
4	Welding specification WS-01 and welder identification available at the work	WS-01, SW-WPS-3	Specification on site; every welder identified	Welder list	R	R	R				
5	Fit-up of every new weld: root opening and alignment within the root face	SW-WPS-3, SW-WD-4	Root opening 1/8 in; bores aligned within 1/16 in	Fit-up record per weld	H	W	S				
6	Preheat and interpass temperatures maintained during welding	WS-01, SW-WPS-2	Preheat ≥ 30 °F above dew point and ≥ 60 °F, 3 in each side; interpass ≤ 450 °F	Weld record per weld (welder, temperatures)	S	S	S				
7	Visual examination of every new weld	SW-NDE-1, SW-NDE-4	No crack, lack of fusion or incomplete penetration	VT report	W	W	W				
8	Volumetric examination per the NDE table; repairs re-examined	SW-NDE-2, SW-NDE-4	Acceptance per SW-NDE-4 and the owner's criteria	RT/UT reports	R	H	R				
9	Flanged ends: faces inspected, new gaskets, studs to preload	SW-NDE-5, SW-WO-5	No radial scratch; preload as stated	Bolt-up record	W	W	—				
10	Hydrostatic test at 210 psi, 30 minutes	SW-TEST-1	No leak, no pressure drop	Test record with gauge calibration	H	H	H				
11	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
12	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

Where the work is on the owner's drawings (SW-IMP-8)

Each affected item, replaced weld and isolation point, with the drawing and sheet that hold it and the weld label printed there. The sheets follow, copied unchanged.

Tag	Role	Drawing	Sheet	Weld label
4"-P150-CWS-3003/p8	affected	ISO-CWS-3003	1	
4"-P150-CWS-3003/el9	affected	ISO-CWS-3003	1	
4"-P150-CWS-3003/p10	affected	ISO-CWS-3003	1	
4"-P150-CWS-3003/el11	affected	ISO-CWS-3003	1	
4"-P150-CWS-3003/p12	affected	ISO-CWS-3003	1	
4"-P150-CWS-3003/W2	replaced by WO-300-06-W01	ISO-CWS-3003	1	W2
4"-P150-CWS-3003/W3	replaced by WO-300-06-W02	ISO-CWS-3003	1	W3
4"-P150-CWS-3003/W4	replaced by WO-300-06-W03	ISO-CWS-3003	1	W4
4"-P150-CWS-3003/W5	replaced by WO-300-06-W04	ISO-CWS-3003	1	W5
4"-P150-CWS-3003/W6	replaced by WO-300-06-W05	ISO-CWS-3003	1	W6
4"-P150-CWS-3003/W7	replaced by WO-300-06-W06	ISO-CWS-3003	1	W7
4"-P150-CWS-3003/A3	isolation: valve	ISO-CWS-3003	1	
4"-P150-CWS-3003/A6	isolation: blind	ISO-CWS-3003	1	
4"-P150-CWS-3003/B2	isolation: valve	ISO-CWS-3003	1	
4"-P150-CWS-3003/B5	isolation: blind	ISO-CWS-3003	1	



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	3	4"	FLANGE WELD-NECK 4" P150 RAISED FACE
2	4	4"	GASKET 4" P150 COMPRESSED SHEET
3	2	4"	VALVE, GATE, WEDGE, OS&Y 4" P150
4	4	4"	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150
5	39'-7 5/16"	4"	PIPE 4" P150 BEVELLED ENDS
6	2	4"	ELBOW 90° R = 5/3 OD 4" P150
7	1	4"	HANGER FOR 4" LINE (HANGER-4x70.0)

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

COMPANY

SMARTWELD

TITLE

WATER (COOLING AND PROCESS)

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. FLANGE BOLT HOLES STRADDLE CENTRELINES

TEST PRESS.	INSULATION	PAINT
-	NONE	-

PWHT	NDE	FLUID
-	-	CWS

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
4"-P150-CWS-3003	W150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-CWS-3003	1 OF 1	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS

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